FORGERON

5-Axis CNC · AI-Operated · Built in Bamako

PITCH PREPARATION

METI Japan & Embassy Visit

UniPods Bootcamp · Addis Ababa · 11 Nov 2026

KEY NUMBERS — MEMORISE THESE

$15K Machine cost (vs $40K+ imported)	**38** AI tools (machine functions)	**5-axis** CNC degrees of freedom	**16** UI languages supported
342 Automated tests	**3h35** R20 dome machined (validation)	**2** Team members (Lamine + Aboubacar)	**0** Specialist programmer needed

OPENING LINE — SAY THIS TO THE METI VISITORS

> *"Japan leads the world in machine tools and industrial robotics. We built this machine in Mali, inspired by that tradition of precision. We want to be the bridge that brings that standard of manufacturing to African workshops — at a price an African workshop can actually afford."*
>
> Why it works: METI is Japan's ministry of industry. Machine tools are Japan's pride. You are not asking for help — you are acknowledging shared values.

5-MINUTE SCRIPT — TIMED

TIME	BEAT	WHAT YOU SAY
0:00	**The problem**	*"In Mali, a workshop that receives an order for a complex precision part has two choices: turn it down, or import it. The import takes 3 weeks and costs 3 times more. That is why complex manufacturing does not exist here."*
1:00	**The machine**	*"Forgeron is a 5-axis CNC machine — it carves a metal part from a solid block. We designed it and built it in Bamako, for $15,000. An equivalent imported machine costs $40,000 and requires a specialist programmer. We removed both barriers."*
2:00	**The AI — DEMO**	*"The operator does not write code. They talk to the machine, in plain language. Watch."* 👉 Open the app. Say a command. The agent explains what it will do. Ask the audience to confirm. The machine moves. 👉 If the machine is not running: show the app screen + AI response. That is enough.
3:30	**Safety principle**	*"The rule in our system is: the AI proposes, the deterministic code calculates, the operator confirms, and the hardware stops. The AI never moves the machine without permission. This is not optional — it is in the architecture."*
4:00	**Vision + Japan**	*"Japan built the world standard for machine tools. We are building the African version — affordable, AI-operated, and in local languages. Our next step is a paid pilot in two workshops in Bamako. We are open to any connection in the precision manufacturing space."*
4:45	**Close**	*"The machine that made complex parts impossible to build in Mali — we built it in Mali."* Show the Africa map piece if machined. Pause. Done.

DEMO FLOW — STEP BY STEP

1. Open the Flutter app on the tablet. Show the home screen to the audience.
   Say: "This is what the workshop owner sees."
2. Type or say a command in English (or French): *"Mill a 20mm dome on aluminium"*
   The agent responds in natural language, describes the toolpath, asks for confirmation.
3. Tap Confirm. The machine starts (or the simulation plays if no live cut).
   Point out: "No G-code. No programmer. The operator pressed one button."
4. Show the Africa map piece (if machined). Hand it to a visitor to touch.
   "This was made on this machine, in this room." Physical proof beats any slide.

LIKELY QUESTIONS & ANSWERS

Q: What is the precision / tolerance of the machine?

> We achieve ±0.1 mm on aluminium in current testing. For harder materials we are still characterising. That level of precision is sufficient for most workshop parts — spare parts, moulds, structural brackets.

Q: What materials can it cut?

> Currently aluminium, brass, wood, and soft plastics. Steel is next — it requires higher rigidity in the frame, which is in our roadmap. **5-axis kinematics are already there; it is a mechanical upgrade.**

Q: How does the AI work — is it connected to the internet?

> It has two modes. Online: Gemini API for maximum capability. Offline: a local model running on the laptop with Ollama — no internet needed. **The safety layer is always local and always active regardless of mode.**

Q: Who are your customers today?

> We do not have paying customers yet. That is the goal of this bootcamp: to close a paid pilot with two workshops in Bamako by April 2027. We have identified the workshops; we are validating the price they will pay.

Q: What is your business model?

> Machine sale to workshops, plus a software subscription for the AI agent. We are also exploring a pay-per-part model for smaller workshops that cannot afford the upfront cost.

Q: How did you build this in Mali — where did you source the components?

> Electronic components from Aliexpress and local suppliers. Structural parts machined locally where possible, some imported from China. The design is entirely ours. **The constraint of sourcing locally forced better design decisions.**

Q: Are you open to partnerships — with Japanese companies?

> Absolutely. Japan leads in precision machine tools — companies like Fanuc, Mazak, Okuma. We see two opportunities: component supply at better terms, and technology transfer for the next version of the machine. We are also open to joint market development for African industry.

Q: What is the market size?

> Africa has hundreds of thousands of small workshops that currently cannot access precision manufacturing. Even 1% of machine shops in West Africa adopting a tool like this is a significant market. The unmet need is real — we see it every day around us in Bamako.

Q: What about intellectual property — is the software open source?

> The core machine design and AI agent are proprietary. We are adding a license this week. We may open-source some components in the future to build an ecosystem, but the core remains protected.

Q: Can the AI make mistakes that damage the machine or hurt someone?

> This was our first design constraint. The AI never moves the machine without operator confirmation. The hardware emergency stop is always active and cannot be gated by software. The rule in the system: **"The AI proposes, the deterministic code calculates, the operator confirms, the hardware stops."**

Q: You mentioned local languages — why does that matter for a CNC machine?

> Most machinists in Mali speak Bambara, not French or English. If the machine speaks only English, the operator who runs it every day is excluded. We are building Bambara support so the person at the machine — not just the engineer who designed it — can operate it in their own language. That is what real adoption looks like.